Oliver Atherton

Huddersfield, U.K. (Open to Relocation) | 07956555742 | oliverhatherton@gmail.com
oliverhatherton.com | linkedin.com/in/oliverhatherton | github.com/oliverhatherton

Education

University of Huddersfield — First-Class Honours (Predicted) West Yorkshire, UK
BSc (Hons) Software Engineering *Sep 2023 – Jul 2027*

- Year 2: 74.2% | Placement year: 84%
- Selected modules: Operating Systems (91%), Software Design (81%), Computer Architecture (79%)

Experience

Software Engineer (Year in Industry) Sep 2025 – Sep 2026
Babcock International Group *Bristol, UK*

- Selected as 1 of 3 software engineering interns from 140+ applicants; delivered production software in a security-controlled, classified environment
- Graded **Outstanding** (highest band) in the final placement appraisal, with work assessed as on par with or better than the company's graduate engineers

Radio Command & Control System — *Java, Spring Boot, Vue, Agile/Scrum*

- Designed the consolidated endpoint contract for asset configuration, specifying behaviour up front so frontend and backend could run in parallel; reduced 16 REST endpoints across 4 hardware types to 8 (13-point ticket, team of three)
- Became productive in the Spring Boot codebase within 3 days; delivered an end-to-end feature (UI, backend, DSL) in 4 days, approximately 50% ahead of estimate
- Refactored the integration test helper, replacing multiple methods and explicit null-parameter passing with overloaded PUT/GET helpers, reducing duplication across the suite
- Ran a technical spike into hardware simulator capabilities and produced a rapid prototype to gather stakeholder feedback ahead of supporting a new hardware type

Monitoring & Configuration Desktop Application — *Rust, Tokio, Tauri, React, Python*

- Reverse-engineered encrypted packet captures in Wireshark to derive undocumented protocol behaviour, unblocking a feature that had previously stalled; documented the method for the team
- Returned a production desktop application, offline for nearly a year, to operational status within 2 weeks
- Reduced application packaging time from ∼45 minutes to ∼5 minutes by externalising sensitive configuration to a runtime-loaded file rather than embedding it in the binary
- Extended a Python SNMP simulator with additional endpoints and corrected device addressing so multiple simulated devices returned distinct data, enabling development without lab hardware

Projects

Distributed Order Processing Platform order-lifecycle.oliverhatherton.com | Code
NestJS, TypeORM, PostgreSQL, RabbitMQ, Redis, Docker, React

- Designed, built and deployed an event-driven order-fulfilment platform (NestJS API, React UI) with an asynchronous PENDING → RESERVED → PAID → COMPLETED pipeline over RabbitMQ
- Guaranteed exactly-once payment processing by pairing a producer-side outbox with a consumer-side inbox (deduplication, bounded retries, dead-lettering), preventing duplicate charges when messages are retried or redelivered
- Secured user sessions against token theft using JWT authentication (in-memory access tokens, rotating httpOnly refresh tokens) with refresh-token reuse detection that revokes compromised session families
- Reduced p99 latency by ∼51% and increased throughput by ∼54% under load with a Redis cache-aside layer over hot reads, measured with k6 load tests against cached and uncached endpoints
- Instrumented the API, message consumers and database with Prometheus, OpenTelemetry and Grafana; backed by 90+ Testcontainers end-to-end tests and 30+ unit suites in GitHub Actions CI, deployed on Docker/Render

Leadership Experience

Backend Lead – Convoy Management System Oct 2025
Babcock Internal Hackathon — 2nd place, team of 4

- Led backend development for a convoy-planning platform (Node.js, Express, Sequelize, SQLite) computing optimal convoy composition from cargo weight, route distance and vehicle range; built and demonstrated live within 72 hours
- Built and maintained the majority of the REST API for vehicles, routes and convoys, and integrated single-click PDF report generation (PDFKit) with the team's Flutter frontend
- Contributed the majority of commits, owned the database schema and onboarded a junior developer

Technical Skills

Languages: Java, TypeScript, JavaScript, SQL, Python, Rust
Backend & Systems: Spring Boot, NestJS, Node.js/Express, REST APIs, RabbitMQ, event-driven architecture
Data & Infrastructure: PostgreSQL, Redis, TypeORM, Docker, GitHub Actions CI/CD, Cloudflare Workers
Testing & Observability: JUnit, Jest, Testcontainers, k6, Prometheus, Grafana, OpenTelemetry
Frontend: React, Vue, Vite